

Cluster Algorithms

DANA 4840

```
#clValid(obj, nClust, clMethods = "hierarchical",
# validation = "stability", maxitems = 600,
# metric = "euclidean", method = "average")
```

- **obj**: A numeric matrix or data frame. Rows are the items to be clustered and columns are samples.

```
library(c1Valid)
# Iris data set:
# - Remove Species column and scale
df <- scale(iris[, -5])
```

Internal Measures

```
# Compute clValid
clmethods <- c("hierarchical", "kmeans", "pam")

intern <- clValid(df, nClust = 2:6,
                  clMethods = clmethods, validation = "internal")

## Warning in clValid(df, nClust = 2:6, clMethods = clmethods, validation = 
## "internal"): rownames for data not specified, using 1:nrow(data)

# Summary
summary(intern)

##
## Clustering Methods:
## hierarchical kmeans pam
##
## Cluster sizes:
## 2 3 4 5 6
##
## Validation Measures:
##
##           2           3           4           5           6
## hierarchical Connectivity 0.9762  5.5964  7.5492 18.0508 24.7306
##                   Dunn    0.2674  0.1874  0.2060  0.0700  0.0762
##                   Silhouette 0.5818  0.4803  0.4067  0.3746  0.3248
## kmeans Connectivity 0.9762 23.8151 25.9044 40.3060 40.1385
##                   Dunn    0.2674  0.0265  0.0700  0.0808  0.0808
##                   Silhouette 0.5818  0.4599  0.4189  0.3455  0.3441
## pam Connectivity 0.9762 23.0726 31.8067 35.7964 44.5413
##                   Dunn    0.2674  0.0571  0.0566  0.0642  0.0361
##                   Silhouette 0.5818  0.4566  0.4091  0.3574  0.3400
##
```

```
## Optimal Scores:
##
##           Score Method      Clusters
## Connectivity 0.9762 hierarchical 2
## Dunn         0.2674 hierarchical 2
## Silhouette   0.5818 hierarchical 2
```

It can be seen that hierarchical clustering with two clusters performs the best in each case (i.e., for connectivity, Dunn and Silhouette measures). Regardless of the clustering algorithm, the optimal number of clusters seems to be two using the three measures.

Stability Measures

```
# Stability measures
clmethods <- c("hierarchical", "kmeans", "pam")
stab <- clValid(df, nClust = 2:6, clMethods = clmethods,
               validation = "stability")

## Warning in clValid(df, nClust = 2:6, clMethods = clmethods, validation =
## "stability"): rownames for data not specified, using 1:nrow(data)

# Display only optimal Scores
optimalScores(stab)

##           Score      Method Clusters
## APN 0.003266667 hierarchical      2
## AD  1.004288856          pam      6
## ADM 0.016087089 hierarchical      2
## FOM 0.455750052          pam      6
```

For the APN and ADM measures, hierarchical clustering with two clusters again gives the best score. For the other measures, PAM with six clusters has the best score.